

NORMAL-FORM COMPLEXITY OF A THREE-SHEETED KELLER COVER

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ABSTRACT. Let $K = X + Q + C$ be a degree-at-most-three Keller map in dimension n , with Q, C homogeneous of degrees two and three. If the coordinate span of C has dimension r , we give a collision-preserving cubic-homogeneous suspension in dimension $n + r + 1$. Applied to the public 11-variable degree-three descendant of the new Jacobian counterexample, where $r = 7$, this recovers the 19-variable construction independently announced by Harris Chan.

We determine the generic Jordan type of its nonlinear Jacobian as $(18, 1)$ and give a two-point certificate that its nilpotent flag moves. Symmetric doubling gives an explicit Hessian-nilpotent quartic in 38 variables, with generic Hessian type $(35, 2, 1)$. A two-parameter family of algebraic inverse rays shows

$$\Delta^m(\mathcal{Q}^{m+1}) \neq 0 \quad \text{for every } m \geq 1,$$

and identifies the inverse singularities with intersections with the non-properness hypersurface. For the fixed 19-variable cubic tensor, we construct a 110-variable square-zero cubic-linear pairing and prove that every vector-Waring decomposition uses at least 52 cubes. Finally, an exact 13-term functional is a constant nonzero quartic obstruction across a natural 20-dimensional equivariant family of rank-six cubic compressions. The example-specific statements are computer-assisted and come with exact rational certificates.

Working computer-assisted manuscript. The recovered exact certificates have been rerun. Independent CAS reproduction and specialist review remain submission requirements.

1. INTRODUCTION

The classical reductions of Bass–Connell–Wright and Yagzhev make cubic-homogeneous maps central to the Jacobian conjecture [3]. Once a three-dimensional counterexample was announced [1], it therefore became natural to ask how economically that counterexample reaches the classical normal forms. We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the example, and Alpöge announced it publicly [10].

Atkins-Turkish publicly supplied a degree-at-most-three map in 11 variables, with three explicit colliding points [2]. Its cubic coordinate space

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has dimension seven. The construction in section 2 consequently gives a cubic-homogeneous counterexample in

$$11 + 7 + 1 = 19$$

variables. Harris Chan independently announced this special sparse lift on July 23, 2026 [4]; we make no sole-priority claim for the 19-variable endpoint. The reusable contribution is the rank-sensitive $n + r + 1$ theorem.

The fixed 19-variable tensor has further structure. Its nonlinear Jacobian has generic Jordan type $(18, 1)$. Classical symmetric reduction [5] then produces a quartic potential in 38 variables. A 48-variable explicit Hessian witness was already public through the work of Santib   ez-Leal, starting from Thompson’s 24-variable cubic map [9, 8]. Our claims are the quantitative dimension 38, the exact Hessian Jordan type $(35, 2, 1)$, and the all-order Laplacian formula.

In the cubic-linear direction, we use the pairing formalism of Gorni and Zampieri [7]. The existence of such pairings and the square-zero mechanism are not new. For this particular tensor we prove the model-relative interval

$$52 \leq N_{\text{pair}} \leq 110.$$

Neither endpoint is asserted to be sharp for arbitrary cubic-homogeneous or cubic-linear counterexamples.

The main results are summarized as follows.

Theorem 1.1 (Rank-sensitive suspension). *Let $K = X + Q + C: \mathbb{A}^n \rightarrow \mathbb{A}^n$ be Keller over a characteristic-zero field, with Q quadratic homogeneous and C cubic homogeneous. Put*

$$r = \dim \text{span} \{C_1, \dots, C_n\}.$$

Then K admits a cubic-homogeneous Keller suspension in dimension $n+r+1$. Every collision of K lifts explicitly to a collision of the suspension.

Theorem 1.2 (The fixed 19-variable tensor). *For the tensor H of section 3:*

- (i) *$I + H$ is a noninjective cubic-homogeneous Keller map in 19 variables;*
- (ii) *the generic Jordan type of JH is $(18, 1)$;*
- (iii) *the associated 38-variable quartic $\mathcal{Q}$ is Hessian nilpotent, has generic Hessian type $(35, 2, 1)$, and violates Zhao’s eventual vanishing conclusion at every order; and*
- (iv) *the vector-Waring length of H is at least 52, while H has a full-rank square-zero pairing of length 110.*

Theorem 1.3 (Equivariant compression obstruction). *Let V be the space of quadratic source fields that preserve the weights; it has dimension 115. The locus in V killing five specified cubic output rows contains a natural affine slice of dimension 20. Each point on that slice has at most six linearly independent cubic coordinate functions. The same functional, supported on 13 terms, takes the value 1 on its quartic obstruction. Thus no point lifts*

through quartic order in the stated equivariant conjugacy model. The original 605-shift statement is a sparse corollary.

Theorem 1.3 is deliberately model-relative. It does not prove that 19 variables are globally minimal.

2. A RANK-SENSITIVE CUBIC SUSPENSION

Let k be a field of characteristic zero and write

$$K(X) = X + Q(X) + C(X),$$

where Q and C are homogeneous of degrees two and three. Choose a basis $q = (q_1, \dots, q_r)^T$ of the coordinate span of C , and a constant matrix $B \in M_{n \times r}(k)$ such that

$$C = Bq.$$

Introduce $w = (w_1, \dots, w_r)$ and define

$$\widetilde{K}(X, w) = (X + Q(X) + Bw, w - q(X)). \quad (2.1)$$

Lemma 2.1. *The map $\widetilde{K}$ is polynomially stably left-right equivalent to K .*

Proof. Set

$$S(X, w) = (X, w - q(X)), \quad T(U, V) = (U + BV, V).$$

Both are triangular polynomial automorphisms, and direct substitution gives

$$\widetilde{K} = T \circ (K \times \text{id}_{\mathbb{A}^r}) \circ S.$$

□

Adjoin a homogenizing coordinate t , and put

$$G(X, w, t) = (X + tQ(X) + t^2Bw, w - q(X), t). \quad (2.2)$$

Its nonlinear part is cubic homogeneous.

Proof of theorem 1.1. The Jacobian matrix in the X, w coordinates has block form

$$\begin{pmatrix} I + tJQ(X) & t^2B \\ -Jq(X) & I \end{pmatrix}.$$

The Schur complement gives

$$\begin{aligned} \det JG &= \det(I + tJQ(X) + t^2BJq(X)) \\ &= \det JK(tX) = 1. \end{aligned} \quad (2.3)$$

If $K(p) = K(p')$, then

$$\widehat{p} = (p, q(p), 1), \quad \widehat{p}' = (p', q(p'), 1)$$

are distinct and satisfy

$$G(\widehat{p}) = (K(p), 0, 1) = G(\widehat{p}').$$

Finally, write $G = I + H$. Since H is cubic homogeneous,

$$JH(sZ) = s^2JH(Z).$$

The identity $\det(I + JH) = 1$ therefore implies $\det(I + \lambda JH) = 1$ as a polynomial identity in λ . All nonconstant characteristic coefficients vanish, so JH is nilpotent. $\square$

Remark 2.2. For a fixed K , any factorization $C = B'q'$ through s scalar cubic coordinates has $s \geq r$. Thus r auxiliaries are optimal within the factorized suspension (2.1). This says nothing about other stable representatives of K or other homogenization models.

3. THE FIXED 19-VARIABLE TENSOR

We now work over $\mathbb{Q}$. In the coordinate order

$$X = (x, y, z, a, b, c, d, q, s, h, k),$$

let $\Phi = (\Phi_1, \dots, \Phi_{11})$ be the public map [2]:

$$\begin{aligned} \Phi_1 &= -ac - adz - 3ay^2 - 2az - cd^2 + d^2z - ds + 7dy^2 \\ &\quad + sxy + 3xyz + 4y^2 + z, \\ \Phi_2 &= -bc - bdz - 3by^2 - 2bz - 3cdx - dq + qxy \\ &\quad + 12xy^2 + 3xz + y, \\ \Phi_3 &= -hk - hxz + kx^2 - 3x^2y + 2x, \\ \Phi_4 &= a - d^2 + 2dxy, \quad \Phi_5 = b + 3x^2y, \\ \Phi_6 &= c + xyz + 3y^2 + 2z, \quad \Phi_7 = d - xy, \\ \Phi_8 &= bz + 3cx + q, \\ \Phi_9 &= s + az + cxy - xyz - 7y^2 + cd - dz, \\ \Phi_{10} &= h - x^2, \quad \Phi_{11} = k + xz. \end{aligned}$$

Its linear part $L = J\Phi(0)$ has determinant -2 . Normalize by

$$K = L^{-1}\Phi = X + Q + C.$$

The coordinate cubics

$$q_1 = C_1, \quad q_2 = C_2, \quad q_3 = C_3, \quad q_4 = C_4, \quad q_5 = C_5, \quad q_6 = C_6, \quad q_7 = C_9$$

are linearly independent, and the remaining coordinate cubics vanish. Thus $C = Bq$, where B includes these seven coordinates into $\mathbb{Q}^{11}$.

With $w = (w_1, \dots, w_7)$, define

$$H(X, w, t) = (tQ(X) + t^2Bw, -q(X), 0) \tag{3.1}$$

in the coordinate order

$$Z = (x, y, z, a, b, c, d, q, s, h, k, w_1, \dots, w_7, t).$$

Then $G = I + H$ is the suspension (2.2).

For reference, three colliding points of Φ are

$$\begin{aligned} p_1 &= (0, 0, -1/4, 0, 0, 1/2, 0, 0, 0, 0), \\ p_2 &= (1, -3/2, 13/2, -9/4, 9/2, -10, -3/2, 3/4, -153/8, 1, -13/2), \\ p_3 &= (-1, 3/2, 13/2, -9/4, -9/2, -10, -3/2, -3/4, -153/8, 1, 13/2). \end{aligned}$$

They all map to $(-1/4, 0, \dots, 0)$. Consequently

$$(p_j, q(p_j), 1), \quad j = 1, 2, 3,$$

are three colliding points of G .

3.1. Generic Jordan type. Let $A = JH$, and set

$$E = xy(c + 2z) + d^2z + 3dy^2.$$

Define two polynomial vectors

$$v_1 = 2tx e_h + 2tz e_k + (xk + zh)e_{w_1}, \quad (3.2)$$

$$v_2 = -txy e_a - t(dz + 3y^2)e_s - Ee_{w_3} + 2Ee_{w_6} + xyz e_{w_7}. \quad (3.3)$$

Proposition 3.1. *Over $\mathbb{Q}(Z)$, the matrix A has Jordan type $(18, 1)$.*

Proof. Exact differentiation gives

$$Av_1 = Av_2 = 0.$$

The vectors are generically independent, while the all-ones specialization of A has rank 17. Hence the generic kernel has dimension two.

Let e_t be the last coordinate vector. Exact iteration gives

$$A^{17}e_t = 18t^{15}x^9y^5z^2v_2 \neq 0. \quad (3.4)$$

By theorem 1.1, A is nilpotent. A nilpotent 19-by-19 matrix with a two-dimensional kernel has two Jordan blocks, and (3.4) forces the larger block to have size at least 18. Thus the partition is $(18, 1)$. $\square$

Remark 3.2. At the all-ones specialization the ranks of $A, A^2, \dots, A^{18}$ are

$$17, 16, \dots, 1, 0.$$

The vectors

$$e_t, Ae_t, \dots, A^{17}e_t, v_1$$

specialize to a basis with determinant 1952152956156672.

4. A RESTRICTED COMPRESSION OBSTRUCTION

Consider quadratic near-identity source conjugacies. At cubic order their action is

$$C \longmapsto C + [Q, P_2], \quad [Q, P_2] = JQ P_2 - JP_2 Q.$$

The shift

$$P_2 = -d^2e_a$$

lowers the cubic coordinate-span rank from seven to six. Its exact conjugate has quartic remainder

$$(D_4)_z = d^3z + 3d^2y^2, \quad (D_4)_a = x^2y^2, \quad (D_4)_c = -2d^3z - 6d^2y^2. \quad (4.1)$$

For a quartic vector field R , define

$$\begin{aligned} \Lambda_4(R) = & [x^2y^2]R_a + 4[x^2ac]R_a - \frac{20}{3}[x^2as]R_a + 9[x^2qk]R_a \\ & + 3[xchk]R_a - \frac{1}{2}[xd^2q]R_a - 7[xshk]R_a + \frac{4}{3}[y^2ah]R_a \\ & - \frac{1}{2}[ybch]R_a - [ydhk]R_a + \frac{8}{3}[za^2h]R_a + 25[aqhk]R_a - 2[dqhk]R_d. \end{aligned} \quad (4.2)$$

Proposition 4.1. *For every cubic vector field P_3 ,*

$$\Lambda_4([Q, P_3]) = 0, \quad \Lambda_4(D_4) = 1.$$

Thus $D_4 \notin \text{im}[Q, -]$.

Moreover, among the 605 shifts

$$P_2 = \lambda m e_j,$$

where m is a quadratic monomial not involving X_j , the shift $-d^2e_a$ is the unique one that lowers the cubic span to six.

Proof. The first statement is finite coefficient linear algebra. Evaluating (4.2) on all

$$11 \binom{13}{3} = 3146$$

cubic monomial vector fields gives zero, while direct evaluation on (4.1) gives one. The restricted homological matrix on the 13 displayed coefficient positions has rank 12 and augmented rank 13.

For the second statement, a fixed nonzero seven-by-seven minor of the cubic coefficient matrix is evaluated along each of the 605 directions. Of these, 583 have no exceptional parameter and 22 give 27 rational roots. Exact full-rank evaluation at every root leaves only $P_2 = -d^2e_a$. $\square$

Proposition 4.2. *Use the quadratic output corrections generated by the four outputs d, q, h, k , which have degree at most two after the conjugacy. Their ten quadratic products create a ten-dimensional quartic image. The z - and c -components of D_4 lie outside this image. The a -component has the unique preimage $-Y_d^2$, but applying it restores cubic coordinate-span rank seven.*

Proof. This is another exact coefficient-matrix calculation. The ten quartic images are independent. Solving against the three nonzero components in (4.1) gives the assertions. $\square$

5. THE 38-VARIABLE HESSIAN QUARTIC

Let $F(x) = x + H(x)$ be the 19-variable map. On $\mathbb{A}_x^{19} \times \mathbb{A}_y^{19}$, define the cotangent double

$$\Phi_F(x, y) = (F(x), JF(x)^T y).$$

Its Jacobian is block lower triangular, with diagonal blocks JF and JF^T , so $\det J\Phi_F = 1$.

Over $\mathbb{Q}(i)$, put

$$T(u, v) = \left(u + iv, \frac{u - iv}{2} \right).$$

The scalar block satisfies

$$TT^T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore

$$\mathcal{G} = T^{-1}\Phi_F T = I + \nabla \mathcal{Q},$$

where

$$\mathcal{Q}(u, v) = \frac{1}{2}(u - iv)^T H(u + iv). \quad (5.1)$$

This is a homogeneous quartic with 724 nonzero monomials. If $F(p) = F(p')$, then

$$\Phi_F(p, 0) = \Phi_F(p', 0),$$

so $\mathcal{G}$ has an explicit transported collision.

Proposition 5.1. *The generic Jordan type of $\text{Hess } \mathcal{Q}$ is*

$$(35, 2, 1).$$

Proof. Before the linear twist, the nonlinear Jacobian is

$$N = \begin{pmatrix} A & 0 \\ \mathcal{L} & A^T \end{pmatrix}, \quad A = JH(x), \quad \mathcal{L} = \text{Hess}_x(y^T H(x)).$$

The Hessian of $\mathcal{Q}$ is similar to N .

The kernel vectors (3.2)–(3.3) satisfy

$$D^2 H[v_1, v_2] = D^2 H[v_2, v_2] = 0.$$

Since they span $\ker A$ generically, this implies $\mathcal{L}v_2 \in \text{im } A^T$. In addition to the two kernel vectors $(0, \ker A^T)$, the matrix N therefore has a kernel vector whose first component is v_2 . Hence $\text{rank } N \leq 35$. An exact all-ones specialization has rank 35.

For $k \geq 1$, the lower-left block of N^k is

$$\sum_{j=0}^{k-1} (A^T)^j \mathcal{L} A^{k-1-j}.$$

Since $A^{18} = 0$, the only possible term for $k = 35$ is $(A^T)^{17} \mathcal{L} A^{17}$. The image of A^{17} is spanned by v_2 , and $D^2 H[v_2, v_2] = 0$, so this term vanishes. Thus $N^{35} = 0$. At the all-ones specialization,

$$(N^{34})_{38,19} = 648 \neq 0.$$

The nilpotency index is 35 and the rank is 35, so there are three blocks. The remaining three dimensions split as $2 + 1$, proving the result. $\square$

5.1. An all-order Laplacian certificate. Let

$$\bar{Y} = (1/2, 1, 0, \dots, 0, 1) \in \mathbb{Q}^{19}, \quad \bar{W} = T^{-1}(\bar{Y}, 0) = \left(\frac{\bar{Y}}{2}, -\frac{i\bar{Y}}{2} \right).$$

An exact inverse branch of the original three-dimensional map, lifted through the stable reductions, gives

$$(F^{-1}(s\bar{Y}))_1 = \frac{1/\sqrt{1-s^4}-1}{s^3} = \sum_{r \geq 1} \frac{1}{4^r} \binom{2r}{r} s^{4r-3}. \quad (5.2)$$

Consequently

$$(\mathcal{G}^{-1}(s\bar{W}))_{u_1} = \frac{1}{2} \sum_{r \geq 1} \frac{1}{4^r} \binom{2r}{r} s^{4r-3}. \quad (5.3)$$

Put $P = -\mathcal{Q}$, so that $\mathcal{G} = I - \nabla P$. Zhao's inversion formula for a Hessian-nilpotent potential gives homogeneous inversion terms [12]

$$R_m = \frac{1}{2^m m! (m+1)!} \Delta^m(P^{m+1}).$$

The degree of ∇R_m is $2m + 3$. Comparing (5.3) at $m = 2r - 3$ gives:

Theorem 5.2. *For every $r \geq 2$,*

$$\partial_{u_1} \Delta^{2r-3}(\mathcal{Q}^{2r-2})(\bar{W}) = \frac{(2r-3)!(2r-2)!}{16} \binom{2r}{r} \neq 0.$$

In particular,

$$\partial_{u_1} \Delta(\mathcal{Q}^2)(\bar{W}) = \frac{3}{4}.$$

Proof. The exponent $m + 1 = 2r - 2$ is even, so $P^{m+1} = \mathcal{Q}^{m+1}$. Multiply the coefficient of s^{4r-3} in (5.3) by $2^m m! (m+1)!$. $\square$

Since $\mathcal{Q}$ is Hessian nilpotent, $\Delta^m(\mathcal{Q}^m) = 0$ for all $m \geq 1$, while theorem 5.2 gives infinitely many nonzero $\Delta^m(\mathcal{Q}^{m+1})$. This is an explicit failure of eventual vanishing in Zhao's framework.

6. A 52-TO-110 SQUARE-ZERO PAIRING INTERVAL

Let $V = \mathbb{Q}^{19}$, and regard

$$H \in V \otimes \text{Sym}^3(V^*).$$

Its *vector-Waring length* is the least R for which

$$H(z) = \sum_{\nu=1}^R b_\nu \ell_\nu(z)^3. \quad (6.1)$$

6.1. **The lower bound.** Let

$$W = \text{span} \{ \partial H_i / \partial z_j \} \subset \text{Sym}^2(V^*).$$

Every decomposition (6.1) gives

$$W \subset U := \text{span} \{ \ell_\nu^2 \}.$$

Exact row reduction gives

$$\dim W = 39. \tag{6.2}$$

In the coordinate order of section 3, the 12-dimensional subspace

$$\mathcal{A} = \langle z, c, q, s, k, w_1, \dots, w_7 \rangle$$

is a common zero block:

$$W|_{\mathcal{A}} = 0.$$

The coefficient system for a constant vector v satisfying $JH(z)v = 0$ has rank 19; a specified 19-by-19 minor has determinant 18. Thus the ℓ_ν span V^* , and their restrictions span $\mathcal{A}^*$. Twelve independent restrictions have twelve independent squares. It follows that

$$\dim(U/W) \geq 12, \quad \dim U \geq 51.$$

Proposition 6.1. *Every vector-Waring decomposition of H has length at least 52.*

Proof. Suppose a decomposition has length at most 51. Every preceding inequality is then an equality:

$$R = \dim U = 51, \quad W = \ker(U \rightarrow \text{Sym}^2(\mathcal{A}^*)).$$

Choose a basis $\alpha_1, \dots, \alpha_{12}$ of $\mathcal{A}^*$ among the restricted forms. If $\alpha = \sum c_j \alpha_j$ and α^2 lies in the span of the α_j^2 , then all $c_i c_j$ for $i \neq j$ vanish. Hence every restricted form is proportional to one α_j .

Choose a complement $V = \mathcal{B} \oplus \mathcal{A}$, with $\dim \mathcal{B} = 7$, and let

$$\mathcal{C} = \text{pr}_{\mathcal{B}^* \otimes \mathcal{A}^*}(W).$$

The equality description forces $\mathcal{C}$ to be invariant under the twelve coordinate projectors in the α_j -basis. Therefore its right stabilizer algebra

$$\mathcal{E} = \{ T \in \text{End}(\mathcal{A}^*) : (I \otimes T)\mathcal{C} \subset \mathcal{C} \}$$

contains a conjugate of the full diagonal algebra. Its commutant can then contain no nonzero nilpotent.

Exact rational computation gives

$$\dim \mathcal{C} = 24, \quad \dim \mathcal{E} = 62, \quad \dim \mathcal{E}' = 7,$$

and $\mathcal{E}'$ contains the matrix unit $E_{5,0}$, with indices numbered $0, \dots, 11$. This element is nonzero and square-zero, a contradiction. $\square$

6.2. The 110-variable upper construction. The supplement contains rational matrices

$$B \in M_{19 \times 110}(\mathbb{Q}), \quad D \in M_{110 \times 19}(\mathbb{Q})$$

satisfying

$$\text{rank } B = \text{rank } D = 19, \quad BD = 0, \quad H(z) = B(Dz)^{*3}. \quad (6.3)$$

Put $A = DB$ and

$$F_A(W) = W + (AW)^{*3}.$$

Then $A^2 = 0$ and $\text{rank } A = 19$.

Proposition 6.2. *The map $F_A: \mathbb{A}^{110} \rightarrow \mathbb{A}^{110}$ is a noninjective Keller map. It is Gorni–Zampieri paired with $G = I + H$.*

Proof. Choose a right inverse C of B . The exact data give $AC = D$, and

$$BF_A(Cz) = z + B(Dz)^{*3} = G(z).$$

By Sylvester’s determinant identity,

$$\begin{aligned} \det JF_A(W) &= \det \left(I_{110} + 3 \text{diag}((AW)^2)DB \right) \\ &= \det \left(I_{19} + 3B \text{diag}((DBW)^2)D \right) \\ &= \det JG(BW) = 1. \end{aligned}$$

If $G(z_1) = G(z_2)$, put $u = Cz_1$, $v = Cz_2$, and $\delta = F_A(u) - F_A(v)$. Then $B\delta = 0$, hence $A\delta = DB\delta = 0$. Thus

$$F_A(v + \delta) = F_A(v) + \delta = F_A(u).$$

The exact collision certificate verifies $u \neq v + \delta$. □

Corollary 6.3. *For the fixed tensor H , the minimum length of a full-rank square-zero pairing satisfies*

$$52 \leq N_{\text{pair}} \leq 110.$$

Proof. The lower bound is proposition 6.1; every such pairing is a vector-Waring decomposition. The upper bound is proposition 6.2. □

7. EXACT COMPUTATION AND REPRODUCIBILITY

All example-specific calculations use exact rational arithmetic. The fresh audit reran:

- (1) the public-input-to-19D collision and nilpotency certificate;
- (2) the rank-seven factorization and Schur-complement identities;
- (3) both kernel vectors, the length-18 Jordan chain, and the degeneration certificate;
- (4) the 38D gradient identity, Hessian rank and index, inverse branch, and first Laplacian value;
- (5) the 110D matrices, square-zero identity, pairing identity, Keller determinant transfer, and collision;

- (6) the 52-cube lower-bound matrices and equality obstruction;
- (7) all 3,146 monomial tests for Λ_4 ;
- (8) all 605 triangular monomial shifts and the ten target-cleanup directions; and
- (9) a separate verifier using only Python rational fractions for the small rank, minor, circuit, and commutant certificates.

The regenerated 724-term quartic is byte-for-byte identical to the recovered output. The 110D pairing JSON and the serialized small certificates also match their archived inputs exactly.

The source, exact serialized tensors, and successful logs are organized in `computational-supplement/`. In particular, `fixed-tensor/` contains the 19D, 38D, and 110D constructions, `continuation/` contains the Waring and compression calculations, and the two `n5-regular-*` directories separate the collision-line existence theorem from the first-normal global obstruction. The claim-to-file map and minimal replay commands are in `COMPUTATION.md`. This organization reflects the mathematical lesson of the calculation: regularity on the collision line is possible, but global nilpotence imposes new normal-direction equations.

The computations do not establish novelty, and the several SymPy scripts are not independent CAS implementations. The Fraction-only checker is independent of SymPy but consumes serialized certificate matrices. A public release should additionally reproduce the main identities in Singular, Macaulay2, Sage, or Magma.

8. QUESTIONS

Question 8.1 (Minimum homogeneous dimension). *What is the minimum dimension of a cubic-homogeneous counterexample? The fixed 19-variable construction gives an upper bound, while the known low-dimensional classifications and the five-dimensional restrictions in appendix A leave a substantial gap.*

Question 8.2. *Can one improve the rank-sensitive complexity*

$$\dim K + \dim \text{span} \{(K_3)_i\}$$

by choosing a different degree-at-most-three stable representative?

Question 8.3 (Intrinsic tensor and pairing ranks). *What are the exact vector-Waring length, pairing rank, and square-zero realization rank of the fixed 19-variable cubic tensor? In particular, can the present vector-Waring interval between 52 and 110 be closed without committing to one reduction algorithm?*

Question 8.4. *Can the $\mathfrak{sl}_{18}$ monolith structure in appendix F be used to derive the Jordan types $(18, 1)$ and $(35, 2, 1)$ without coefficient calculation?*

APPENDIX A. THE FIVE-DIMENSIONAL FRONTIER

The body studies explicit descendants in dimensions 19, 38, and 110. The opposite optimization problem begins in dimension five: cubic-homogeneous Keller maps in dimensions three and four are known to be invertible, so five is the first dimension in which a cubic-homogeneous counterexample could occur. This appendix records exact restrictions and a surviving line-level family. Nothing here constructs a five-dimensional counterexample.

A.1. Collision pencils. Let

$$F(x) = x + H(x), \quad H(x) = T(x, x, x),$$

where $T: V^3 \rightarrow V$ is symmetric and $\dim V = 5$. Suppose

$$F(v + u) = F(v)$$

with u, v linearly independent. On $\mathbb{P}(\langle u, v \rangle)$, write

$$M(s, t) = JH(su + tv) = s^2A + stB + t^2C,$$

where

$$A = JH(u), \quad B = 6T(u, v, -), \quad C = JH(v).$$

Symmetry of T and the collision give

$$Bu = 2Av, \quad Bv = 2Cu, \quad u + \frac{1}{3}Au + Av + Cu = 0. \quad (\text{A.1})$$

Conversely, the first two identities are sufficient to extend the plane data to a symmetric trilinear tensor. They do not ensure that the global Jacobian is nilpotent away from the plane.

Proposition A.1 (Three generic Jordan strata). *For a five-dimensional cubic-homogeneous counterexample, the generic nilpotent Jordan type of JH must be one of*

$$(5), \quad (4, 1), \quad (3, 2).$$

Proof. Known low-rank results exclude generic rank at most two in a five-dimensional counterexample, while nilpotence gives rank at most four. The nilpotent partitions of five having rank three or four are precisely the three displayed partitions. $\square$

A.2. A corrected sparse exclusion. Consider the square-zero two-matrix family

$$H_{C,Q}(z) = C \left((Qz)^{*3} - Qz^{*3} \right).$$

After a signed-permutation normalization, let $Q = I + R$, where R has exactly three active rows and exactly one nonzero off-diagonal entry in each active row. Impose the coordinate collision

$$(I + H_{C,Q})(e_0) = (I + H_{C,Q})(e_1).$$

Theorem A.2 (Three-row sparse exclusion). *No member of this coordinate-collision ansatz whose three directed support edges collectively use all five vertices is Keller.*

Proof. For one edge $i \rightarrow j$ with coefficient r , its cubic contribution is

$$U = (z_i + rz_j)^3 - z_i^3 - rz_j^3.$$

If b is the corresponding column of the compressed coefficient matrix, the z_i^2 and $z_i z_j$ coefficients in its contribution to the trace are

$$3rb_j, \quad 6rb_i + 6r^2b_j.$$

Thus the trace equation forces $b_i = b_j = 0$.

There are 180 three-edge patterns with distinct source rows. The collision and trace constraints leave 30, forming three orbits under the evident vertex symmetries:

$$\begin{array}{l|l} A & (0 \rightarrow 2, 3 \rightarrow 0, 4 \rightarrow 1), \\ B & (2 \rightarrow 0, 3 \rightarrow 0, 4 \rightarrow 1), \\ C & (2 \rightarrow 0, 3 \rightarrow 1, 4 \rightarrow 2). \end{array}$$

Writing the edge coefficients as α, β, γ , exact elimination of the collision and trace equations leaves, respectively, the following nonzero coefficients in $\text{tr}(JH/3)^2$:

$$\frac{18}{\beta^2 - 1} \quad \text{in orbit } A, \quad 18 \quad \text{in orbit } C.$$

In the generic branch of orbit B , two coefficients force both

$$B_{1,1} = 0, \quad (\beta^3 - \beta)B_{1,1} = 1.$$

The branches $\alpha = \pm 1$ and $\beta = \pm 1$ instead leave

$$\frac{36\beta}{\beta^2 - 1} \quad \text{or} \quad \frac{36\alpha}{\alpha^2 - 1}.$$

Every branch is contradictory. $\square$

Remark A.3 (Scope correction). The theorem is not the blanket statement that every support- ≤ 4 five-dimensional ansatz fails. Its coordinate-collision, three-row, and one-edge-per-row hypotheses are essential. An earlier finite-field experiment using only coefficients $0, \pm 1$ was vacuous for this collision and is not evidence for the theorem; the exact characteristic-zero argument above is the proof.

A.3. Everywhere-regular Jordan type (5). Assume $M(s, t)$ has Jordan type (5) at every point of $\mathbb{P}^1$. Regard it as

$$M: \mathcal{O}_{\mathbb{P}^1}^5 \longrightarrow \mathcal{O}_{\mathbb{P}^1}^5(2).$$

For $K_i = \ker M^i$ and $L_i = K_i/K_{i-1}$, the induced maps

$$L_i \longrightarrow L_{i-1}(2)$$

are isomorphisms. Since the degrees sum to zero,

$$(\deg L_1, \dots, \deg L_5) = (-4, -2, 0, 2, 4). \quad (\text{A.2})$$

In particular, the right kernel is an $\mathcal{O}(-4)$ -subbundle of $\mathcal{O}^5$.

If r is the coordinate-span dimension of a quartic kernel generator, the quadratic-syzygy calculation leaves only

r	possible splitting
3	(2, 2)
4	(1, 1, 2)
5	(1, 1, 1, 1),

while $r = 2$ is impossible.

In the full-span chart, normalize the kernel to the rational normal quartic

$$k(s, t) = (s^4, s^3t, s^2t^2, st^3, t^4)^T.$$

Its linear syzygy matrix is

$$K(s, t) = \begin{pmatrix} -t & s & 0 & 0 & 0 \\ 0 & -t & s & 0 & 0 \\ 0 & 0 & -t & s & 0 \\ 0 & 0 & 0 & -t & s \end{pmatrix}.$$

The linear lifting system for the regular nilpotent quotient has rank 48 in 56 unknowns and hence an eight-dimensional solution space. On the open set where the resulting constant 4×4 matrix T is invertible, the pencil factors as

$$M(s, t) = R(s, t)T^{-1}K(s, t)$$

and satisfies $M^5 = 0$.

Theorem A.4 (A smooth characteristic-zero collision-line family). *The everywhere-regular type-(5) equations (A.1) have a smooth one-dimensional family in characteristic zero.*

Exact computer-assisted existence proof. In the eight-parameter full-kernel chart, the following point over $\mathbb{F}_{11}$ satisfies the integrability and collision equations:

$$\begin{aligned} a &= (8, 7, 1, 7, 2, 9, 0, 1), \\ u &= (7, 6, 3, 10, 4), \quad v = (8, 9, 7, 9, 1). \end{aligned}$$

At this point $\det T = 1$. The gcd of the 4×4 minors of $M(s, t)$ is one in $\mathbb{F}_{11}[s, t]$, so the pencil is geometrically regular on $\mathbb{P}_{\mathbb{F}_{11}}^1$.

After fixing $a_7 = 1$ and one coordinate of v , the ten integrability and five collision equations are functions of 16 variables. Their Jacobian has rank 15; a specified 15×15 minor has determinant 3 modulo 11. The multivariate Hensel lemma therefore gives a one-dimensional smooth characteristic-zero lift. The two accompanying dimension-five supplement directories contain independent exact Hensel replay scripts. They check the lift through modulus

$$11^{12} = 3138428376721.$$

Geometric regularity is open, so it persists on the lifted family. $\square$

A.4. The first-normal obstruction. The smooth family in theorem A.4 is a family of collision-line data. It does not extend to a global nilpotent Jacobian. Choose coordinates with $u = e_0$, $v = e_1$, and let n_0, n_1, n_2 span the normal directions. For each r , put

$$N_r(s, t) = \frac{d}{d\epsilon} JH(su + tv + \epsilon n_r) \Big|_{\epsilon=0} = sP_r + tQ_r.$$

Tensor symmetry fixes

$$P_r u = 2An_r, \quad P_r v = Bn_r, \quad Q_r u = Bn_r, \quad Q_r v = 2Cn_r.$$

The remaining first-normal tensor entries form a 60-dimensional vector space. Nilpotence through first normal order requires

$$\text{tr}(M(s, t)^{k-1} N_r(s, t)) = 0, \quad r = 0, 1, 2, \quad k = 1, \dots, 5. \quad (\text{A.3})$$

Theorem A.5 (First-normal obstruction on the smooth residue disk). *The complete smooth $\mathbb{Z}_{11}$ -residue disk through the point in theorem A.4 has no extension through first normal order to a symmetric cubic tensor with nilpotent Jacobian.*

In collision-adapted coordinates, the equations for $k \leq 4$ in (A.3) have a 60×60 coefficient matrix with determinant $3 \pmod{11}$. Adding $k = 5$ gives

$$\text{rank } L = 60, \quad \text{rank}[L \mid b] = 61,$$

with a selected augmented 61×61 minor of determinant $1 \pmod{11}$.

Exact computer-assisted proof. At the displayed residue point, the transformed line matrices are

$$A = \begin{pmatrix} 6 & 4 & 0 & 10 & 2 \\ 6 & 1 & 0 & 9 & 10 \\ 8 & 0 & 0 & 6 & 2 \\ 3 & 8 & 0 & 5 & 2 \\ 3 & 6 & 0 & 1 & 10 \end{pmatrix},$$

$$B = \begin{pmatrix} 8 & 8 & 1 & 4 & 7 \\ 2 & 5 & 2 & 6 & 9 \\ 0 & 2 & 5 & 4 & 4 \\ 5 & 4 & 6 & 9 & 7 \\ 1 & 8 & 10 & 3 & 6 \end{pmatrix}, \quad C = \begin{pmatrix} 4 & 0 & 5 & 1 & 3 \\ 8 & 1 & 6 & 6 & 6 \\ 1 & 8 & 5 & 6 & 10 \\ 2 & 6 & 0 & 0 & 2 \\ 4 & 7 & 9 & 1 & 1 \end{pmatrix}.$$

The first four trace identities form a square system in the 60 first-normal coefficients and have determinant 3. Substituting its unique solution into the fifth identities leaves the coefficient matrix

$$\begin{pmatrix} 5 & 8 & 9 & 1 & 9 & 2 & 2 & 1 & 3 & 1 \\ 2 & 2 & 6 & 2 & 4 & 6 & 1 & 7 & 2 & 5 \\ 9 & 5 & 9 & 8 & 5 & 3 & 4 & 6 & 3 & 8 \end{pmatrix},$$

whose first three columns have determinant 2. This proves inconsistency. The determinants 3 and 1 are 11-adic units, so the contradiction persists throughout the residue disk.

An independent coordinate-free calculation starts with all

$$5 \binom{7}{3} = 175$$

coefficients of a symmetric vector-valued cubic. The line restrictions have rank 65; adjoining the identities through $k = 4$ gives coefficient and augmented ranks $(125, 125)$; adjoining $k = 5$ gives $(125, 126)$. The remaining 50 variables after $k \leq 4$ are exactly the pure-normal coefficients $5 \binom{5}{3}$, which cannot affect first normal order. This confirms that the obstruction is independent of the chosen normal complement. $\square$

The same exact test was applied exhaustively to the rational points of the eight-parameter full-kernel chart:

	$\mathbb{F}_7$	$\mathbb{F}_{11}$
projective parameter points	960800	21435888
invertible chart matrix	806344	19324668
regular collision-line solutions	6	11
first-normal extensions	0	0.

Every surviving pencil is geometrically regular over the algebraic closure, and every first-normal system has ranks $(60, 61)$. These enumerations exclude the rational points in the displayed chart and their good-reduction local branches; they do not classify points over finite extensions or other kernel-span strata.

Question A.6 (Global first-normal obstruction). *On the regular full-kernel collision-line curve, the first-normal incompatibility defines a covariant*

$$\Omega \in N_P^* \otimes H^0(\mathbb{P}^1, \mathcal{O}(9)).$$

Is Ω nowhere zero? An affirmative answer would exclude the entire full-kernel regular type-(5) stratum, not only the displayed residue disk and the rational points over $\mathbb{F}_7, \mathbb{F}_{11}$.

A.5. A global rank-drop obstruction. Let $M(x)$ be a homogeneous quadratic nilpotent 5×5 matrix on $\mathbb{P}^4$, generically of type (5), and put

$$D = \{[x] \in \mathbb{P}^4 : \text{rank } M(x) \leq 3\}.$$

Theorem A.7 (Codimension-two rank drop). *One has*

$$\text{codim}_{\mathbb{P}^4} D \leq 2.$$

If M extends an everywhere-regular collision line, then

$$\text{codim}_{\mathbb{P}^4} D = 2$$

and D is disjoint from that line.

Proof. Suppose $\text{codim } D \geq 3$ and put $U = \mathbb{P}^4 \setminus D$. The regular kernel filtration on U has line quotients

$$L_i \simeq \mathcal{O}_U(-4 + 2(i - 1))$$

by the same calculation as (A.2). Removing a codimension-at-least-three set preserves Pic and CH^2 . Multiplicativity of Chern classes in the filtration of $\mathcal{O}_U^5$ would give

$$0 = c_2(\mathcal{O}_U^5) = \sum_{0 \leq i < j \leq 4} (-4 + 2i)(-4 + 2j)h^2 = -20h^2,$$

contradicting $h^2 \neq 0$ in $\text{CH}^2(U)$.

An everywhere-regular collision line is disjoint from D . A divisor in $\mathbb{P}^4$ cannot be disjoint from a projective line, so D cannot have codimension one. The preceding bound then forces codimension two. $\square$

Remark A.8 (The remaining problem). Theorem A.5 eliminates the explicit smooth full-kernel residue disk, but not every regular type-(5) collision-line component. The kernel-span-three and kernel-span-four strata and the generic Jordan types (4, 1) and (3, 2) also remain open. Every successful global extension must acquire the codimension-two rank-drop surface of theorem A.7. Thus dimension five remains possible, but the known regular line family is not the restriction of a counterexample.

APPENDIX B. ONE COVER IN SEVERAL NORMAL FORMS

The reductions below do more than transport a collision: after adjoining purely transcendental variables, they preserve the function-field extension induced by the original map. Thus the cubic, symmetric, and square-zero descendants are presentations of the same generically three-sheeted cover.

Proposition B.1 (Rees presentation of the rank-sensitive suspension). *Let $K = X + Q + C$, let $C = Bq$, and let*

$$G(X, w, t) = (X + tQ(X) + t^2Bw, w - q(X), t).$$

Define triangular automorphisms

$$S(X, v, t) = (X, v + q(X), t), \quad T(Y, v, t) = (Y - t^2Bv, v, t).$$

Then

$$T \circ G \circ S = (K_t \times \text{id})(X, v, t), \quad K_t(X) = X + tQ(X) + t^2C(X) = t^{-1}K(tX).$$

Consequently, after inverting t , the function-field extension induced by G is obtained from that induced by K by adjoining independent transcendental variables. In particular, generic degree and geometric monodromy are preserved.

Proof. Direct substitution gives

$$G(X, v + q(X), t) = (X + tQ(X) + t^2Bv + t^2C(X), v, t),$$

and the target shear T removes t^2Bv . Over $k(t)$ the change of variables $U = tX$ identifies K_t with K . $\square$

Proposition B.2 (The symmetric double is a stable presentation). *Let $F = X + H$ be Keller and put*

$$\Phi_F(x, y) = (F(x), JF(x)^T y).$$

Then the polynomial automorphism

$$S_F(x, y) = (x, JF(x)^T y)$$

satisfies

$$\Phi_F = (F \times \text{id}) \circ S_F.$$

Hence the symmetric double, and therefore its linear gradient twist, induces the same function-field extension as F after adjoining $\dim F$ independent variables.

Proof. Since $\det JF = 1$, the inverse matrix JF^{-T} has polynomial entries, so S_F is a polynomial automorphism. $\square$

Proposition B.3 (A full-rank square-zero pairing is stably right-equivalent). *Suppose*

$$H(z) = B(Dz)^{*3}, \quad BD = 0,$$

*where $B : W \rightarrow V$ is surjective and $D : V \rightarrow W$ is injective. Put $A = DB$ and $F_A(w) = w + (Aw)^{*3}$. Choose a right inverse C of B and write $W = C(V) \oplus E$, where $E = \ker B$. Set*

$$\rho(z) = (Dz)^{*3} - CH(z) \in E.$$

In the coordinates $w = Cz + \eta$, one has

$$F_A(z, \eta) = (G(z), \eta + \rho(z)), \quad G = I + H.$$

Precomposition by $(z, \eta) \mapsto (z, \eta - \rho(z))$ therefore changes F_A into $G \times \text{id}_E$.

B.1. The quotient cubic and its monodromy. For the original three-variable map, write its target coordinates as (ξ, η, ζ) . The source has the grading $\text{wt}(x, y, z) = (1, -1, -2)$, and the target weights are $(-2, -1, 1)$. Put

$$\nu = xy, \quad v = x^2z, \quad r = 1 + \nu, \quad c = 2 - 3\nu - v, \quad q = cr,$$

and form the target invariants

$$A = \xi\zeta^2, \quad B = \eta\zeta.$$

A direct calculation gives

$$A = q^2 + cq - q^3, \quad B = 4q + 2c - 3q^2.$$

Eliminating c yields the cubic

$$q^3 - 2q^2 + Bq - 2A = 0. \tag{B.1}$$

Its discriminant is

$$-4(27A^2 - 18AB + 16A + B^3 - B^2) = -4\zeta^2 \mathcal{N}(\xi, \eta, \zeta),$$

where

$$\mathcal{N}(\xi, \eta, \zeta) = 27\xi^2\zeta^2 - 18\xi\eta\zeta + 16\xi + \eta^3\zeta - \eta^2. \quad (\text{B.2})$$

The quadratic polynomial in A inside the discriminant has discriminant $-4(3B - 4)^3$, hence is not a square over $\mathbb{Q}(B)$. Together with the generic irreducibility of (B.1), this gives geometric monodromy S_3 .

APPENDIX C. A TWO-PARAMETER FAMILY OF ALL-ORDER INVERSE RAYS

Let e_x, e_y, e_{w_1}, e_t refer to the indicated coordinates of the 19-variable cubic map, and for $\alpha, \beta \in \mathbb{Q}$ put

$$Y_{\alpha, \beta} = \alpha e_x + e_y + \beta e_{w_1} + e_t.$$

Let $\widehat{G}_0^{-1}$ denote the formal inverse branch at the origin.

Theorem C.1 (Exact inverse-ray family). *For all α, β ,*

$$(\widehat{G}_0^{-1}(sY_{\alpha, \beta}))_x = \frac{(1 - 2\alpha s^4 + 2\beta s^6)^{-1/2} - 1}{s^3},$$

where the square root is the branch with constant term 1.

Proof. The t -coordinate gives $t = s$, and the w -coordinates give $w = q(X) + s\beta e_1$. Therefore

$$K(sX) = s^2(\alpha e_x + e_y) - s^4\beta e_x.$$

After applying the linear normalization and eliminating the stable auxiliary coordinates, this is the original three-variable target curve

$$(\xi, \eta, \zeta) = (0, s^2, 2\alpha s^2 - 2\beta s^4).$$

On the local branch above $\xi = 0$, write

$$R = (1 + xy)^2 z + y^2(4 + 3xy).$$

The equation $\xi = (1 + xy)R = 0$ gives $R = 0$, and then $\eta = y = s^2$. With $r = 1 + xy$, the third component simplifies to

$$\zeta = \frac{1 - r^{-2}}{y}.$$

Thus

$$r = (1 - y\zeta)^{-1/2} = (1 - 2\alpha s^4 + 2\beta s^6)^{-1/2},$$

and the original x -coordinate is $(r - 1)/s^2$. Since the original source is sX , division by s gives the formula. $\square$

The same calculation explains the singularities: along this curve,

$$\mathcal{N}(0, s^2, 2\alpha s^2 - 2\beta s^4) = -s^4(1 - 2\alpha s^4 + 2\beta s^6).$$

Thus the algebraic singularities of the inverse ray are precisely its intersections with the discriminant/nonproperness hypersurface.

Let $\mathcal{G} = I + \nabla Q$ be the 38-variable gradient descendant, and put

$$W_{\alpha, \beta} = T^{-1}(Y_{\alpha, \beta}, 0).$$

For $n \geq 0$ define

$$c_n(\alpha, \beta) = [u^n](1 - 2\alpha u^2 - 2\beta u^3)^{-1/2}.$$

For $n \geq 2$ one has the finite sum

$$c_n(\alpha, \beta) = \sum_{k=\lceil n/3 \rceil}^{\lfloor n/2 \rfloor} 2^{-k} \binom{2k}{k} \binom{k}{n-2k} \alpha^{3k-n} \beta^{n-2k}. \quad (\text{C.1})$$

In particular, $c_n(\alpha, \beta) > 0$ for positive rational α, β .

Theorem C.2 (Nonvanishing at every Laplacian order). *For every $m \geq 1$ and every positive rational α, β ,*

$$\partial_{u_1} \Delta^m(\mathcal{Q}^{m+1})(W_{\alpha, \beta}) = 2^{m-1} m!(m+1)! c_{m+3}(\alpha, \beta) > 0.$$

Thus the same quartic violates the eventual-vanishing conclusion at every order, not merely along an infinite parity class.

Proof. The first coordinate of the inverse gradient ray is half the first coordinate in Theorem C.1. If

$$(1 - 2\alpha u^2 + 2\beta u^3)^{-1/2} = \sum d_n u^n,$$

then $d_n = (-1)^n c_n(\alpha, \beta)$. Compare the coefficient of s^{2m+3} with Zhao's inversion formula for $P = -\mathcal{Q}$:

$$R_m = \frac{1}{2^m m!(m+1)!} \Delta^m(P^{m+1}).$$

The two signs cancel, giving the displayed positive formula. $\square$

The central-binomial formula in the earlier draft is the boundary case $(\alpha, \beta) = (1/2, 0)$. Turning on β fills the parity gaps.

APPENDIX D. THE NILPOTENT FLAG MOVES

Let $A(Z) = JH(Z)$ for the 19-variable cubic tensor.

Proposition D.1 (A two-point obstruction to strong nilpotence). *Although $A(Z)$ is nilpotent for every Z , the family is not strongly nilpotent and is not simultaneously strictly triangularizable. Indeed,*

$$A(e_d)A(e_t)|_{\langle e_{w_3}, e_{w_6} \rangle} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix},$$

and all other entries of the product vanish. Hence

$$\chi_{A(e_d)A(e_t)}(\lambda) = \lambda^{18}(\lambda + 3).$$

This gives a concrete form of the distinction between pointwise nilpotence and a common nilpotent flag: the generic matrix is almost regular nilpotent, but its Jordan flag twists with the base point.

D.1. A self-dual extension lemma.

Lemma D.2 (Isotropic terminal vector). *Let A have Jordan type $(m, 1)$, let $\ker A = \langle v_1, v_2 \rangle$, and suppose $\operatorname{im} A^{m-1} = \langle v_2 \rangle$. Let $L = L^T$ and*

$$N = \begin{pmatrix} A & 0 \\ L & A^T \end{pmatrix}.$$

Define the symmetric form

$$\beta(u, v) = u^T L v \quad (u, v \in \ker A).$$

If β has rank one with radical $\langle v_2 \rangle$, if $N^{2m-2} \neq 0$, and if $\beta(v_2, v_2) = 0$, then N has Jordan type $(2m-1, 2, 1)$.

Proof. Projection of $\ker N$ to its first factor identifies its image with $\operatorname{rad} \beta$, and each fiber is $\ker A^T$. Thus $\dim \ker N = 2 + 1 = 3$. The lower-left block of N^{2m-1} is the sole possible term

$$(A^T)^{m-1} L A^{m-1},$$

which vanishes because the terminal line is isotropic. The preceding power does not vanish:

$$N^{2m-2} \neq 0.$$

Thus the nilpotency index is $2m-1$. The operator acts on a space of dimension $2m+2$ and has three blocks. Its largest block has size $2m-1$, so its partition is $(2m-1, 2, 1)$. $\square$

For the present tensor, exact differentiation gives

$$D^2 H[v_1, v_1] = -4t^3 x z e_x, \quad D^2 H[v_1, v_2] = D^2 H[v_2, v_2] = 0.$$

Hence the form induced by $L = \operatorname{Hess}_x(y^T H(x))$ has rank one on $\ker A$, with radical $\langle v_2 \rangle$. The previously verified nonzero entry of N^{34} then applies Lemma D.2 with $m = 18$ and explains the partition $(35, 2, 1)$ as a one-box degeneration of maximal chain gluing.

APPENDIX E. A ROBUST EQUIVARIANT COMPRESSION OBSTRUCTION

The normalized 11-variable map is equivariant for the weights

$$(1, -1, -2, 0, 1, -2, 0, -1, -2, 2, -1)$$

on $(x, y, z, a, b, c, d, q, s, h, k)$. Let $\mathfrak{X}_2^{(0)}$ be the space of weight-preserving quadratic vector fields; exact counting gives $\dim \mathfrak{X}_2^{(0)} = 115$.

For a quadratic formal coordinate change $I + P$, the cubic jet is $C + [Q, P]$. Its quartic term is

$$\mathcal{O}_4(P) = [C, P] + \frac{1}{2} D^2 Q[P, P] - J P[Q, P] - \frac{1}{2} D^2 P[Q, Q]. \quad (\text{E.1})$$

Put

$$\Sigma_{-2} = \langle y^2, qy, ky, az, dz, ac, as, cd, ds, q^2, kq, k^2 \rangle$$

and

$$P_\sigma = -d^2 e_a + \sigma e_s \quad (\sigma \in \Sigma_{-2}).$$

Proposition E.1 (A positive-dimensional obstructed compression family). *For every $\sigma \in \Sigma_{-2}$, the cubic jet $C + [Q, P_\sigma]$ is supported only in the six output coordinates x, y, z, b, c, s , and therefore has coordinate-span rank at most six. Nevertheless,*

$$\Lambda_4(\mathcal{O}_4(P_\sigma)) = 1.$$

Hence no cubic source correction removes the quartic term anywhere in this 12-parameter formal family. Excluding the two monomials involving s gives a 10-parameter family of actual triangular polynomial automorphisms.

A larger exact calculation gives the following natural slice. Let

$$\begin{aligned} \mathcal{S} = \{P \in \mathfrak{X}_2^{(0)} : (C + [Q, P])_a = (C + [Q, P])_d = (C + [Q, P])_q = 0, \\ (C + [Q, P])_h = (C + [Q, P])_k = 0\}. \end{aligned}$$

Then $\mathcal{S}$ is a nonempty affine space of dimension 20, and

$$\Lambda_4(\mathcal{O}_4(P)) = 1 \quad (P \in \mathcal{S}).$$

The tangent space at $P_0 = -d^2 e_a$ to the equivariant rank-at-most-six locus has dimension 22; the two extra directions are precisely the infinitesimal weight-preserving target changes that add the surviving y -row to the q - and k -rows. Thus $\mathcal{S}$ is the expected 20-dimensional transverse slice modulo linear target reparametrization.

This changes the interpretation of the obstruction: cubic compression is not isolated or difficult. It occurs in a family, while a secondary quartic class is constant and nonzero across the family.

APPENDIX F. COLLISION MONOLITHS AND PROLONGATIONS

The operator calculations for the fixed tensor suggest a structural theorem for any dimension-minimal cubic-homogeneous counterexample. We formulate the ternary product by

$$F(x) = x + \mu(x, x, x).$$

If $p = a + d$ and $q = a - d$ collide, then

$$d + 3\mu(a, a, d) + \mu(d, d, d) = 0. \tag{F.1}$$

Proposition F.1 (The collision monolith). *Assume that F has minimum dimension among noninjective cubic-homogeneous Keller maps. Its ternary algebra may be replaced by the subalgebra generated by a, d , and then:*

- (i) $d \in A^3$ and $\dim A/A^3 \leq 1$;
- (ii) every nonzero proper ideal contains d ;
- (iii) either A is simple or it has a unique minimal nonzero ideal D , containing d ;
- (iv) $\mu(D, D, D) \neq 0$; and
- (v) if $n = \dim A$, then

$$\dim D \geq \left\lceil \frac{n-1}{2} \right\rceil.$$

Proof. The generated subalgebra retains the collision and the Engel condition, so minimality makes it all of A . Equation (F.1) puts d in A^3 ; modulo A^3 , only the generator a can remain. A proper ideal not containing d would leave a smaller collision in the quotient, proving the monolith assertion.

If $D \neq A$, split $A = B \oplus D$. The map is triangular over its quotient, and the collision lies in one Keller fiber $f_b: D \rightarrow D$ of degree at most three. If $\mu(D, D, D) = 0$, that fiber is quadratic, contradicting Wang's quadratic Keller theorem [11]. If $c = \dim \mu(D, D, D)$, the rank-sensitive suspension applied to f_b would give a cubic-homogeneous counterexample in dimension $\dim D + c + 1 \leq 2 \dim D + 1$. Minimality gives the stated bound. $\square$

Let

$$\mathfrak{g}_D = \text{Lie}\langle L_{x,y}|_D : x, y \in A \rangle \subseteq \mathfrak{gl}(D).$$

The monolith is irreducible under these partial multiplications. The Engel condition makes every $L_{x,x}$, and hence every polarized $L_{x,y}$, traceless. Thus $\mathfrak{g}_D$ is irreducible, semisimple, and contained in $\mathfrak{sl}(D)$.

Theorem F.2 (Monolith–prolongation dichotomy). *After extending scalars to an algebraic closure,*

$$\mathfrak{g}_D = \mathfrak{sl}(D) \quad \text{or} \quad \mathfrak{g}_D = \mathfrak{sp}(D, \omega).$$

Proof. By proposition F.1, the nonzero symmetric tensor $\mu|_{\text{Sym}^3 D}$ belongs to

$$\mathfrak{g}_D^{(2)} = \text{Hom}(\text{Sym}^3 D, D) \cap \text{Hom}(\text{Sym}^2 D, \mathfrak{g}_D).$$

The Cartan–Kobayashi–Nagano classification, in this prolongation convention, says that an irreducible linear Lie algebra with nonzero second prolongation is one of $\mathfrak{gl}$, $\mathfrak{sl}$, $\mathfrak{sp}$, or $\mathfrak{csp}$ [6, Thm. 1.1 and Def. 2.1]. Semisimplicity and tracelessness remove the two conformal cases. $\square$

If $B = A/D$, tensor symmetry puts the D -valued layers in

$$\mathfrak{g}_D^{(2)}, \quad \mathfrak{g}_D^{(1)} \otimes B^*, \quad \mathfrak{g}_D \otimes \text{Sym}^2 B^*, \quad D \otimes \text{Sym}^3 B^*.$$

Every dimension-minimal counterexample is therefore a special-linear or symplectic prolongation extension, rather than an arbitrary ternary algebra.

F.1. The fixed 18 + 1 tensor.

Proposition F.3 (The fixed special-linear core). *For the displayed 19-variable tensor, write $A = U \oplus \mathbb{Q}t$, with $\dim U = 18$, and let W be the seven-dimensional auxiliary-output space. Exact calculation gives*

$$A^3 = U, \quad U^3 = W, \quad \mu(W, U, U) = 0.$$

The partial-multiplication layers generate the full associative algebra

$$\text{End}(U).$$

The restricted Lie algebra is

$$\mathfrak{g}_U = \mathfrak{sl}_{18}.$$

Indeed, irreducibility and theorem F.2 leave only the special-linear and symplectic cases. The equations

$$M_i^T J + J M_i = 0, \quad J^T = -J,$$

for three collision Jacobians form a 153-unknown system of full rank 153 modulo 1,000,003, excluding an invariant symplectic form.

The affine collision Jacobians generate $\mathfrak{sl}_{18} \ltimes \mathbb{Q}^{18}$, of dimension 341, and their unital associative closure has dimension 342. The scalar series has a nonzero 19×19 Hankel minor. With the corrected odd-arity density argument, the ternary algebra fails the 19-variable Capelli system and satisfies the 20-variable system.

These quantities are presentation-relative. The explicit stable suspension changes the affine Lie and associative dimensions from (29, 30) in the 11-variable degree-three presentation to (341, 342) in the homogeneous presentation. Hence neither these dimensions nor the raw Hankel rank is a stable-cover invariant, and they do not prove dimension minimality.

Three exact scripts replay, respectively, the affine Lie and Hankel calculation, the prolongation structure, and the $\mathfrak{sl}_2$ -collision obstruction. Their filenames, JSON inputs, and corrected derivation note are listed in the computation index.

F.2. The remaining extension problem.

Question F.4 (Extension theorem $E(N)$). *For $r + q \leq N$, can one exclude every extension of a one-generated Yagzhev quotient of dimension q by the standard $\mathfrak{sl}_r$ - or $\mathfrak{sp}_r$ -module that simultaneously satisfies the Engel trace identities, irreducibility, the collision equation, and minimality of the monolith? The first focused case is the codimension-one symplectic branch.*

The statement $E(16)$ would give a lower bound 17, while $E(18)$ would match the present upper bound 19. Neither statement is claimed here.

APPENDIX G. SUPPLEMENTARY RESULTS AND RESEARCH LEADS

This appendix keeps potentially relevant results from the research archive attached to a mathematical home without enlarging the main theorem spine. Inclusion here is deliberately permissive. Each entry states its evidence boundary; entries labelled as open, contextual, or evidence-only are not used as theorems elsewhere in the paper. Earlier formulations known to be false are replaced by their current correction. The computational supplement and its `COMPUTATION.md` index give the available program-level artifact map; an entry here does not turn a shared-code check into an independent verification.

G.1. Additional results and constructions.

An explicit degree-three counterexample in 11 variables. *Status.* Proof offered; independent review is not recorded.

The cited map from complex affine 11-space to itself has total degree 3, 52 nonzero monomial terms, constant Jacobian determinant -2, and three displayed distinct rational inputs with one common image.

Credit. Credit recorded to ChatGPT (research assistance); Spacerat (computation).

An explicit cubic-homogeneous counterexample in 24 variables. *Status.* Proof offered; independent review is not recorded.

The cited explicit map has the form $G(U)=U+H(U)$ over $\mathbb{Q}^{24}$, with every nonzero component of H homogeneous cubic, determinant 1, 54 nonzero cubic monomials, and a displayed collision of two distinct rational points.

Credit. Credit recorded to William Thompson (computation).

Starting from the public 11D counterexample, rank-sensitive cubic...

Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

Starting from the public 11D counterexample, rank-sensitive cubic suspension yields an explicit 19D cubic-homogeneous Keller counterexample with collision.

An explicit square-zero Druzkowski pairing of the 19D cubic gives... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

An explicit square-zero Druzkowski pairing of the 19D cubic gives a 135D Druzkowski counterexample with collision.

The earlier 26D cubic-homogeneous and 158D Druzkowski constructions...

Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The earlier 26D cubic-homogeneous and 158D Druzkowski constructions remain valid explicit examples but are superseded as upper bounds by the later 19D and 135D examples.

For an n -dimensional degree-at-most-three map whose quadratic span...

Status. Proof offered; independent review is not recorded.

For an n -dimensional degree-at-most-three map whose quadratic span has rank r , a block-determinant suspension produces a cubic-homogeneous map in dimension $n+r+1$.

For the 19D example, the generic Jacobian perturbation JH has... *Status.*

Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

For the 19D example, the generic Jacobian perturbation JH has Jordan type $(18, 1)$; nonzero scalar fibers are conjugate and the zero fiber is an automorphism degeneration.

The explicit constructions give the upper bound $N_{\min} \leq 19$... *Status.*

Proof offered; independent review is not recorded.

The explicit constructions give the upper bound $N_{\min} \leq 19$ for cubic-homogeneous counterexamples; the conversation uses the elementary no-collinear-collision argument for the lower bound $N_{\min} \geq 5$.

The five-attacks package constructs a 38D Hessian quartic with... *Status*. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The five-attacks package constructs a 38D Hessian quartic with Hessian Jordan type $(35, 2, 1)$ and derives exact central-binomial inverse-ray coefficients with infinitely many nonzero Laplacian terms in the stated parity class.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

The explicit pairing dictionary gives a 110D Druzkowski... *Status*.

Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The explicit pairing dictionary gives a 110D Druzkowski construction, while the derivative-space zero-block and equality-case argument give the model-relative bound $52 \leq \text{pairing rank} \leq 110$ for the fixed 19D tensor. The unique scalar-monomial quadratic shift lowering the cubic jet... *Status*. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The unique scalar-monomial quadratic shift lowering the cubic jet from rank seven to six is $P2 = -d^2 e_a$, and its quartic error is detected by an exact 13-term functional outside the cubic homological image; the natural quadratic target cleanup also fails.

An exact row-defect theorem holds in the two-matrix square-zero... *Status*. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

An exact row-defect theorem holds in the two-matrix square-zero family; the accompanying finite-field search found no better example in the tested range.

A second counterexample $G : \mathbb{C}^4 \rightarrow \mathbb{C}^4$ has... *Status*. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

A second counterexample $G : \mathbb{C}^4 \rightarrow \mathbb{C}^4$ has determinant 4 and generic degree 8.

A straightforward explicit reduction produced a cubic-homogeneous... *Status*. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

A straightforward explicit reduction produced a cubic-homogeneous map in 79 variables and a Druzkowski cubic-linear map in 426 variables.

$MC(SU(79))$ is false. *Status*. Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

$MC(SU(79))$ is false.

The new 24-variable map improves this to $\text{MC}(SU(N))$... *Status.*

Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The new 24-variable map improves this to $\text{MC}(SU(N))$ false for all $N \geq 24$.

The rank-sensitive cubic suspension is triangularly left-right... *Status.*

Proof offered; independent review is not recorded.

The rank-sensitive cubic suspension is triangularly left-right equivalent to (X, v, t) mapped by $(t^{-1}K(tX), v, t)$; it preserves generic degree and the geometric Galois closure after adjoining transcendental variables.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

For any Keller map F , the symmetric double $(F(x), JF(x)^T y)$... *Status.*

Proof offered; independent review is not recorded.

For any Keller map F , the symmetric double $(F(x), JF(x)^T y)$ is polynomially right-equivalent to F times the identity, so the 38-variable gradient descendant has the same generic degree and monodromy as the 19-variable cubic map.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

A full-rank square-zero Gorni-Zampieri pairing $H=B(Dz)^{\wedge 3}$, $BD=0$...

Status. Proof offered; independent review is not recorded.

A full-rank square-zero Gorni-Zampieri pairing $H=B(Dz)^{\wedge 3}$, $BD=0$, is triangularly right-equivalent to the paired cubic map times an identity factor; the 110-variable Druzkowski map is therefore a stable presentation of the 19-variable cubic map.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

The original three-variable map has the displayed invariant quotient...

Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The original three-variable map has the displayed invariant quotient cubic $\theta^3 - 2\theta^2 + B\theta - 2A$, whose irreducibility and nonsquare discriminant give generic Galois group S_3 ; the descendant ladder presents this same three-sheeted cover.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

For the fixed 38-variable Hessian quartic, $\Delta^m(Q^{\wedge m+1})$ is... *Status.*

Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

For the fixed 38-variable Hessian quartic, $\Delta^m(Q^{\wedge m+1})$ is nonzero for every $m \geq 1$; an explicit two-parameter inverse ray removes the earlier parity gap and ties every nonzero coefficient to a sheet escaping at the discriminant.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

Although every $JH(z)$ is nilpotent, the product $JH(e_d)JH(e_t)$ has...

Status. Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

Although every $JH(z)$ is nilpotent, the product $JH(e_d)JH(e_t)$ has characteristic polynomial $\lambda^{18}(\lambda+3)$; hence the polynomial-matrix family is not strongly nilpotent and has no common constant triangularizing basis.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

The equivariant cubic rank-six compression condition has a... *Status.*

Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The equivariant cubic rank-six compression condition has a 20-dimensional affine slice, and the quartic obstruction functional is identically one on that slice; modulo two target-row directions this is the natural transverse compression space.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

The $n+r+1$ cubic suspension is coordinate-free on V plus the... *Status.*

Proof offered; independent review is not recorded.

The $n+r+1$ cubic suspension is coordinate-free on V plus the dual of the output-mode span plus one scalar, and $r=\text{rank}(C^{\text{flat}})$ is the minimum possible auxiliary dimension for any factorization of the cubic tensor through an intermediate vector space.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

Within the fixed 38-dimensional symmetric double, there is no proper...

Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

Within the fixed 38-dimensional symmetric double, there is no proper nondegenerate invariant linear slice surjecting onto the original variables; any smaller realization must change the realization mechanism rather than delete such a slice.

Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).

If an n -dimensional homogeneous map has a k -dimensional... *Status.* Proof offered; independent review is not recorded.

If an n -dimensional homogeneous map has a k -dimensional partial-gradient block, it admits a weak symmetric dilation in dimension at most $2n-k$; in particular every polynomial map has weak symmetric-dilation dimension at most $2n-1$.

Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).

For the displayed 19-dimensional block tensor... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

For the displayed 19-dimensional block tensor $H(X, w, t) = (tQ(X) + t^2Bw, -q(X), 0)$, the weak symmetric-dilation invariant is at most $37-\dim(w)$, hence at most 36.

Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).

The admissible symmetric-overlap index κ gives the lower bound...

Status. Proof offered; independent review is not recorded.

The admissible symmetric-overlap index κ gives the lower bound $\sigma_{\text{sym}}(H) \geq 2n - \kappa_{\text{ov}}(H)$; when the overlap form is nondegenerate, the partial-gradient construction gives the matching upper bound $2n - \kappa_{\text{nd}}(H)$.

Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation). In the scalar-w block case, a 35-dimensional weak dilation extending...

Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

In the scalar-w block case, a 35-dimensional weak dilation extending the evident (w, t) -compression exists exactly when there is a nonzero vector a satisfying $a^T B = 0$, $a^T Q(X) = 0$, and $D_a q(X) = 0$ identically.

Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).

For the principal sl_2 conic on $\text{Sym}^m(k^2)$, the integrability...

Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

For the principal sl_2 conic on $\text{Sym}^m(k^2)$, the integrability operator has determinant $-m^{m+2}(m+2)^m$; no direct sum of principal conics supports a noncollinear normalized collision.

Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).

An everywhere-regular five-dimensional quadratic nilpotent line... *Status.* Proof offered; independent review is not recorded.

An everywhere-regular five-dimensional quadratic nilpotent line pencil whose kernel map spans P^4 has saturated filtration degrees $(-4, -2, 0, 2, 4)$; every successive map is an isomorphism and the kernel map is the rational normal quartic.

Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).

Every quadratic matrix annihilating the Veronese kernel k_m factors...

Status. Proof offered; independent review is not recorded.

Every quadratic matrix annihilating the Veronese kernel k_m factors uniquely as $N = L_m S_m$; iterated Hilbert-Burch descent preserves regular nilpotence, and a fully compatible canonical descent chain is necessarily a common scalar multiple of the principal chain.

Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).

With the quartic kernel fixed at the principal point, the... *Status.*

Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

With the quartic kernel fixed at the principal point, the nilpotent-pencil tangent space has dimension 16 and sl_2 type $V_0 + V_2 + V_4 + V_6$; fixing the descended cubic kernel leaves only scalar rescaling, so the other 15 directions are exactly relative descended-cubic motion.

Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).

For a general five-dimensional quadratic pencil, integrable markings...

Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

For a general five-dimensional quadratic pencil, integrable markings lie on the proper discriminant $\det \Phi_N = 0$, and normalized collisions lie on the determinantal locus $\text{rank } \Theta_N \leq 9$ together with the open independence condition on the recovered marking vectors.

Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).

For a global regular-type five-dimensional nilpotent quadratic... *Status.*

Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

For a global regular-type five-dimensional nilpotent quadratic tensor with full-span collision-line kernel, the saturated rank-one defects are nested and have total codimension-two class $20H^2$, combined Hilbert polynomial $10n^2 + 50n + 81$, and weighted effective increments $5\Delta_1 + 4\Delta_2 + 3\Delta_3 + 2\Delta_4 + \Delta_5 = 20H^2$.

Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).

G.2. Open problems and unresolved assertions.

What is the true smallest dimension of a cubic-homogeneous... *Status.*

Open problem or unresolved assertion.

What is the true smallest dimension of a cubic-homogeneous counterexample?

What is the exact minimal square-zero Druzkowski pairing rank for...

Status. Open problem or unresolved assertion.

What is the exact minimal square-zero Druzkowski pairing rank for the 19D cubic?

G.3. Context, corrections, and evidence boundaries.

Claimed failure of the Mathieu conjecture for $SU(3)$. *Status.* Correction retained: the original positive claim is not adopted; only the displayed negative scope statement is current.

The cited dimension-varying implication does not establish that the Mathieu conjecture for the fixed group $SU(3)$ is false.

Credit. Credit recorded to Zihan Zhang (derivation, exposition).

Within a zero-diagonal five-by-five sparse matrix ansatz with at... *Status.*

Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

Within a zero-diagonal five-by-five sparse matrix ansatz with at most four nonzero entries, no Keller map in the specified H_C family realizes the normalized collision.

A second issue occurs in Zwart's fixed-highest-weight-vector proof. *Status.*

Reported context; not independently established here.

A second issue occurs in Zwart's fixed-highest-weight-vector proof.

No explicit Mathieu witness pair is currently recorded. *Status.* Reported context; not independently established here.

The Mathieu argument currently proves existence of a violating pair, but no explicit closed-form pair of functions on $SU(24)$ or $SU(79)$ is recorded. Making the generic Cartan-twist parameter explicit remains open.

The direct Euclidean 36-dimensional partial-gradient potential is... *Status.* Corrected statement; the earlier stronger formulation is not adopted.

The direct Euclidean 36-dimensional partial-gradient potential is not Keller when B is nonzero; obtaining a 36-dimensional Keller or cover-preserving realization is a separate self-adjoint nilpotent matrix-completion problem.

Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).

AI AND COMPUTATIONAL DISCLOSURE

AI systems were used extensively in exploration, symbolic-program development, proof drafting, source organization, and manuscript editing. They are not authors. The public 11-variable source also discloses that its explicit construction and verification code were generated by ChatGPT [2]. The evidence for the computer-assisted theorems is the exact certificate suite described in section 7, not language model output. Human specialist review has not yet been completed.

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